I discovered that, if Miller's planet is about as near Gargantua as it can get without falling in,[16] and if Gargantua is spinning fast enough, then Chris's one-hour-in-seven-years time slowing is possible. But Gargantua has to spin *awfully fast.*

There is a maximum spin rate that any black hole can have. If it spins faster than that maximum, its horizon disappears, leaving the singularity inside it wide open for all the universe to see; that is, making it *naked*—which is probably forbidden by the laws of physics (Chapter 26).

I found that Chris's huge slowing of time requires Gargantua to spin almost as fast as the maximum: less than the maximum by about *one part in 100 trillion.*[17] In most of my science interpretations of *Interstellar*, I assume this spin.

The crew of the *Endurance* could measure the spin rate directly by watching from far, far away as the robot TARS falls into Gargantua (Figure 6.2).[18] As seen from afar, TARS never crosses the horizon (because signals he sends after crossing can't get out of the black hole). Instead, TARS' infall appears to slow down, and he appears to hover just above the horizon. And as he hovers, Gargantua's whirling space sweeps him around and around Gargantua, as seen from afar. With Garantua's spin very near the maximum possible, TARS' orbital period is about one hour, as seen from afar.

You can do the math yourself: the orbital distance around Gargantua is a billion kilometers and TARS covers that distance in one hour, so his speed as measured from afar is about a billion kilometers per hour, which is approximately the speed of light! If Gargantua were spinning faster than the maximum, TARS would whip around faster than the speed of light, which violates Einstein's speed limit. This is a heuristic way to understand why there is a maximum possible spin for any black hole.